

Mini Project (15%)

Predictive Modelling Using Python

In the era of big data, organizations generate vast amounts of information from customer transactions, operational systems, digital platforms, sensors, and other sources. When analyzed systematically, these data can support better decisions, improve efficiency, identify risks, and reveal opportunities. This mini project requires students to apply a complete data science workflow to a structured dataset using Python in a Jupyter Notebook environment.

The project follows the Data Science Process: Business Understanding, Data Understanding/Collection, Data Preparation, Modelling, Evaluation, and Deployment/Communication. The process is iterative; students may revisit earlier stages when new findings emerge.

Student Brief: Each group will receive one lecturer-provided CSV dataset containing approximately 10,000-15,000 records. The datasets include numerical and categorical attributes and may contain realistic data-quality issues. Your role is to investigate the data, justify how it should be prepared, build and compare predictive models, and communicate actionable findings. Do not assume the dataset is already clean or model-ready.

1. Project Learning Outcomes

- Translate a domain problem into clear analytics objectives and answerable questions.
- Explore a medium-sized structured dataset using appropriate descriptive statistics and visualisations.
- Identify and treat data-quality problems using justified preprocessing decisions.
- Engineer suitable features and prepare model-ready data without introducing data leakage.
- Build, compare, and evaluate multiple predictive models using appropriate metrics.
- Interpret model results, discuss limitations, and communicate evidence-based recommendations.
- Produce reproducible Python analysis in a well-organized Jupyter Notebook.

2. Group Arrangement and Scope

- This is a group task of 2 students.
- Each group will be assigned one dataset and one main prediction problem.
- All analysis must be performed using Python, preferably in Jupyter Notebook.
- Students are expected to demonstrate their own analytical decisions; merely executing pre-written code is insufficient.
- The supplied dataset is the starting point. Cleaning, transformation, encoding, feature engineering, and model preparation form part of the assessment.

3. Dataset Allocation and Main Project Question

The lecturer will provide the CSV file for each group. The following table defines the project domain, prediction target, and primary question. Students may formulate additional supporting questions during Business Understanding.

Group	Dataset	Main Target	Type	Primary Project Question
1	Student Performance Classification	academic_risk	Binary Classification	Predict whether a student is academically At Risk or Not At Risk?
2	Online Retail Purchase	purchase_status	Binary Classification	Predict whether an online retail session results in a Purchase or No Purchase?
3	Employee Attrition	attrition	Binary Classification	Predict whether an employee will Leave or Stay?
4	Property Sale Speed	sale_speed	Binary Classification	Predict whether a listed property will have a Fast Sale or Slow Sale?
5	Food Delivery Delay	delivery_status	Binary Classification	Predict whether a food delivery will be Delayed or On Time?
6	Ecommerce Churn	churn_status	Binary Classification	Predict whether an e-commerce customer will Churn or remain Active?
7	Used Car Demand	market_demand	Binary Classification	Predict whether a used car listing has High Demand or Low Demand?
8	Bank Loan Default	loan_default	Binary Classification	Predict whether a loan applicant will Default or not?
9	Hotel Booking Cancellation	booking_status	Binary Classification	Predict whether a hotel booking will be Cancelled or Not Cancelled?
10	Electricity Consumption Class	consumption_class	Binary Classification	Predict whether a household has High Consumption or Normal Consumption?
11	Telco Customer Churn	customer_status	Binary Classification	Predict whether a telco customer will Churn or Stay?
12	Airline Passenger Satisfaction	airline_satisfaction	Binary Classification	Predict whether a passenger is Satisfied or Dissatisfied?
13	Insurance Claim Risk	claim_risk	Binary Classification	Predict whether an insurance policyholder is High Risk or Low Risk for a claim?
14	Online Learning Completion	course_completion	Binary Classification	Predict whether an online learner will Complete or Not Complete the course?
15	Supermarket Promotion Response	promotion_response	Binary Classification	Predict whether a supermarket customer will Respond or show No Response to a promotion?

4. Required Data Science Process and Report Flow

Your report and notebook should follow the stages below. The examples are general; adapt them to your assigned dataset and prediction type.

Stage	Required Evidence
Business Understanding	- Define the domain problem, prediction objective, and why it matters. - Formulate one main objective and at least 3 supporting analytical/research questions. - Identify who could use the findings and what value the analysis may provide. - State the target variable and whether the task is classification or regression.
Data Understanding	- Load the CSV and report shape, attribute names, data types, and target distribution. - Produce descriptive statistics for numerical and categorical features. - Investigate missing values, duplicates, unusual values, inconsistent categories, and possible outliers. - Use at least 5 meaningful visualisations overall to reveal distributions, comparisons, relationships, trends, or target patterns.
Data Preparation	- Handle missing values and duplicates using justified methods. - Correct or manage invalid values, inconsistent categories, and outliers where appropriate. - Encode categorical variables and transform/scale numerical variables when required by the model. - Create useful derived features where appropriate and explain why they may help. - Split the data into training and testing sets before model fitting; use stratification for classification when appropriate.
Modelling	- Implement at least 3 suitable machine-learning models. - Include a simple/baseline model and at least 2 alternative models for comparison. - Train models only using the training data and document important model parameters. - Perform reasonable tuning or parameter experimentation for at least one model.
Evaluation	- Classification: report confusion matrix plus suitable metrics such as accuracy, precision, recall, F1-score and, where appropriate, ROC-AUC. - Regression: report suitable metrics such as MAE, RMSE and R^2, plus residual/error analysis where appropriate. - Compare all models fairly on the same test set and justify selection of the best model. - Interpret important predictors/features and relate the findings back to the project questions. - Discuss limitations, bias/imbalance concerns, and possible improvements.
Deployment / Communication	Develop a simple interface using Flask, Django, Streamlit, or another suitable Python framework.

Stage	Required Evidence
	• Allow users to enter/select relevant values, run the selected predictive model, and display the prediction result clearly. • Summarise the recommended model and findings in non-technical language. • Explain how the results can support decisions in the assigned domain. • Suggest practical next steps, monitoring considerations, and useful future data. • Include a short demonstration or screenshots of the interface in the report and presentation.

5. Minimum Technical Requirements

- Use Python libraries appropriately, such as pandas, NumPy, matplotlib, scikit-learn and other justified libraries.
- The notebook must run in a logical top-to-bottom sequence without relying on hidden/manual intermediate steps.
- Use a fixed random_state where relevant so that train/test splits and model results are reproducible.
- Do not use the test set to fit scalers, encoders, imputers, feature selectors, or model parameters. Pipelines are encouraged.
- For imbalanced classification data, discuss class balance and use metrics that reflect minority-class performance. Resampling is optional but must be applied only to training data.
- Clearly distinguish raw/original fields from engineered features.
- Every graph must have a meaningful title, labelled axes, and an interpretation in the report—not just an image without discussion.
- Code should include concise comments/Markdown explanations for major analysis steps.

6. Required Deliverables and File Naming

Submit all files in the provided Microsoft teams

Deliverable	Required File Name	Requirements
Presentation slides	PX_slides.pptx	Must contain the major problem statement, data findings, preprocessing decisions, model comparison, conclusion, and recommendations. First slide must list all group members with their pictures.
Original dataset	PX_data_original.csv	The lecturer-provided CSV file, unchanged.
Cleaned/model-ready dataset	PX_data_cleaned.csv	The final cleaned dataset used for analysis/model preparation. If additional experimental datasets are created, name them clearly.
Jupyter Notebook	PX.ipynb	Complete executable analysis with Markdown explanations, visualisations, preprocessing, modelling, evaluation, and conclusion.
Final Report	PX_report.docx	Formal report following the required Data Science Process. Include citations/references for all external sources.

Important: Do not place trained models inside a CSV file. A CSV should contain tabular data only. If you choose to save a trained model as an additional artifact, use a suitable format such as .pkl or .joblib and document how it was created.

7. Final Report Requirements

- Recommended length: approximately **15-20** pages excluding appendices, unless otherwise instructed by the lecturer.
- Use the Data Science Process stages as the main report structure.
- Include relevant tables and figures, but avoid filling the report with raw notebook output.
- Explain the reasoning behind preprocessing and model choices.
- Report both positive and negative findings; do not hide weak model results.
- Include a conclusion that directly answers the project objectives/questions.
- Include properly formatted references for datasets, articles, documentation, and other sources used.
- Large code listings should remain in the notebook rather than being copied extensively into the report.

8. Presentation Requirements

- Presentations will be held in **Week 6 and Week 7**.
- Each group is allocated a maximum of 15 minutes including Q&A.
- Both group members must participate in the presentation and be prepared to answer questions about the analysis and code.
- Focus on insights and analytical decisions rather than reading code line-by-line.
- Show the model comparison clearly and explain why the selected model is appropriate.

- Conclude with key findings, limitations, and practical recommendations.

9. Grading Criteria

Criteria	Description	Weight
Business Understanding	Clarity of problem, objectives, supporting questions, target definition, and domain/business value	10%
Data Understanding & EDA	Completeness of exploration, statistics, data-quality investigation, target analysis, and meaningful visualisation	15%
Data Preparation & Feature Engineering	Correctness and justification of cleaning, missing-value treatment, encoding, transformations, feature engineering, splitting, and leakage prevention	20%
Modelling	Correct implementation of at least 3 suitable models, including a baseline and reasonable parameter experimentation	15%
Evaluation & Interpretation	Appropriate metrics, fair model comparison, justification of best model, feature/result interpretation, limitations, and improvements	15%
Report Quality	Organization, clarity, technical accuracy, visualisations, references, and adherence to the Data Science Process	10%
Presentation	Delivery, clarity, time management, contribution of both members, and ability to answer questions	10%
Teamwork & Submission Compliance	Completeness of deliverables, file naming, reproducibility, academic integrity, and deadline compliance	5%

10. Academic Integrity, References, and Responsible Use of AI

- All submitted analysis must represent the group's own understanding and work. Students must be able to explain any code, model, figure, or conclusion submitted.
- Copying another group's notebook, report, figures, or analysis is not permitted.
- All external articles, code references, documentation, and online resources used in the project must be acknowledged/cited appropriately.
- If generative AI tools are permitted by the lecturer, their use must comply with university/course policy. AI-generated code or text must be checked, understood, tested, and adapted by the students; responsibility for correctness remains with the group.
- Do not fabricate results, citations, experiments, screenshots, or model performance.

11. Submission Checklist

- ☐ Folder is named Project_groupX.
- ☐ Original CSV is included and unchanged.
- ☐ Cleaned/model-ready CSV is included.
- ☐ Jupyter Notebook runs from start to finish and contains explanations.
- ☐ At least 5 meaningful visualisations are included and interpreted.
- ☐ At least 3 suitable predictive models are implemented and compared.
- ☐ Correct evaluation metrics are used for classification or regression.
- ☐ Final report follows the Data Science Process and includes references.
- ☐ Final report is uploaded to Turnitin.
- ☐ Presentation slides include member names/pictures and all key results.
- ☐ All required files are uploaded to the OneDrive folder before the deadline.

Reminder: Marks are awarded for the quality of reasoning and evidence, not simply for obtaining the highest model accuracy. A well-justified analysis that identifies data limitations and evaluates models correctly is stronger than an unexplained high score.